

Staff memo

Mapping Optimal Policy into a Rule in NEMO: GEORG

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Mapping Optimal Policy into a Rule in NEMO: GEORG

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Abstract

This paper documents GEORG, "*Ganske Enkel Optimal ReGel*", a policy rule designed to replicate the loss function based policy rate path in Norges Bank's main macro model, NEMO. The rule is forward looking and uses a selection of macroeconomic indicators of inflation and output developments. Coefficients are estimated by impulse response matching (IRF matching), ie the parameters are set so that the model dynamics of using GEORG in NEMO closely match those of the standard loss function based optimal policy. We outline how the rule can be used to compute a model based policy rate path and how to decompose model based path changes into intuitive forecast revisions.

1 Introduction

Norges Bank’s monetary policy analysis relies on the structural DSGE model, NEMO, in which the policy rate is determined by minimizing a loss function that represents the central bank’s objectives. Specifically, the central bank chooses the policy rate path that minimizes the expected discounted value of the loss over a given horizon subject to the model’s behavioral equations. Weights in the loss function capture the trade-off between output and inflation stabilization, and the resulting model-based policy rate path is considered optimal for the specified weights and model structure. However, the resulting path is only optimal for the specific model parameters and is therefore sensitive to model uncertainty.

Policy rules, linking the policy rate directly to a selection of macroeconomic variables, are more widely used to describe monetary policy behavior. The canonical example is the Taylor rule (Taylor, 1993), which describes how the nominal interest rate should respond systematically to deviations in inflation from its target and output from potential. Although not necessarily optimal, these rules offer a transparent description of monetary policy that can be easily communicated. Extensions and variations of the Taylor rule have since been developed to better fit specific policy environments, incorporating elements such as exchange rate movements or policy rate smoothing to reflect real world policy behavior (Nakamura et al., 2025).

It is well established that effective monetary policy should be forward-looking (Svensson, 2010; Batini and Haldane, 1999) as monetary policy affects the economy with a lag and its impact on GDP and inflation typically peaks after one to two years (Norges Bank, 2024). This dimension can be incorporated into monetary policy rules. Forward-looking rules introduce expectations explicitly, allowing the policy rate to respond to forecasts of inflation, output, and other relevant variables rather than contemporaneous values. In addition to forecasts, such rules can include indicators of inflation and output developments, which improves the responsiveness of policy to underlying tendencies of monetary policy target variables.

In forward-looking New Keynesian frameworks, optimal policy is determined by the expected path of key economic variables (Galí, 2008; Clarida et al., 1999; Woodford, 2002). Therefore, policy rules that incorporate forecasts and predictive indicators are likely better suited to estimate optimal policy than those relying solely on current observations. Along with the policy rate smoothing term, this implies that the policy rate responds more strongly to persistent, medium-term developments and less to short-lived fluctuations and data noise, a response pattern widely regarded as a desirable feature of sound monetary policy (Norges Bank monetary policy strategy, 2025).

To approximate NEMO’s optimal policy using an extended forward looking policy rule, we introduce GEORG, "Ganske Enkel Optimal ReGel",¹ which aims to offer a more transparent complement to the current loss function approach used in NEMO. While the loss function implies that the interest policy rate responds to all variables in the model, GEORG relies directly on a selection of variables relevant to Norwegian monetary policy deliberation: inflation, output gap, unit labor cost, the krone exchange rate, foreign interest rates and the money market premium. The parameters associated with the variables in GEORG are obtained via using an impulse

¹This translates to "A fairly simple optimal rule" in English.

response function (IRF) matched to NEMO’s loss function benchmark.² This approach ensures that NEMO with GEORG replicate the policy response and underlying dynamics of NEMO with the loss function. Targeting the optimal policy differentiates GEORG from conventional policy rules in the literature and from the Taylor type rule employed as a robustness check in the Monetary Policy Report. (([Lindalen and Maffei-Faccioli, 2025](#)))

This Memo documents GEORG and the estimation of the policy rule coefficients. Based on the policy rate paths generated by NEMO with GEORG, we construct a reduced form decomposition analysis illustrating how changes to our macro forecasts have led to adjustments to NEMO’s policy path between reports. This is different from the methodology currently used in Norges Bank’s Monetary Policy Report in which changes to the contributions of exogenous shocks are the basis for the decomposition analysis. Determining a relationship between forecast revisions and exogenous shock decompositions requires deep knowledge of the model’s dynamics. However, this alternative approach relates the decomposition analysis to forecast revisions, which provides a clear mapping from observable forecast changes to the model-based policy rate path.

This Memo is organized as follows. Section 2 describes GEORG in more detail, Section 3 illustrates the use of NEMO with GEORG in practice including policy rate forecasts and decomposition analyses and Section 4 concludes.

²See [Appendix B](#) for a technical description of the estimation.

2 GEORG

This section provides a detailed overview of the GEORG rule and explains how its components are defined and implemented within Norges Bank's forecasting environment. Let r_t denote the policy rate implied by the rule in quarter t , and let $\bar{r}_t$ denote the steady-state nominal policy rate.³ The rule can be expressed as:

$$r_t = \bar{r}_t + \omega_r(r_{t-1} - \bar{r}_t) + (1 - \omega_r)\mathcal{X}_t + Z_t, \quad (1)$$

where $\omega_r \in [0, 1)$ captures the degree of interest policy rate smoothing and the tendency for gradualism in policy settings. Z_t is a residual (monetary policy shock), with its persistence estimated to be consistent with the persistence of the monetary policy shock currently used in NEMO with the loss function approach.⁴ Formally the shock follows an autoregressive process:

$$Z_{t+1} = \lambda_Z Z_t + \varepsilon_{t+1}, \quad |\lambda_Z| < 1. \quad (2)$$

The composite term $\mathcal{X}_t$ collects the macro variables that enter the rule:

$$\mathcal{X}_t = \mathbb{E}_t \left[\omega_\pi \hat{\pi}_{t+1} + \omega_y \hat{y}_{t+1} + \omega_\phi \hat{\phi}_{t+1} + \omega_S \hat{S}_{t+1} + \omega_{rf} \hat{r}_{t+1}^f \right] + \omega_\mu \hat{\mu}_t. \quad (3)$$

Where:

- $\hat{\pi}$: Inflation (CPI-ATE) gap
- $\hat{y}$: Domestic output gap
- $\hat{\phi}$: Unit labor cost growth gap
- $\hat{S}$: Exchange rate growth gap
- $\hat{r}^f$: Foreign policy rate gap
- $\hat{\mu}$: Money market premium gap.

All variables are defined as *gaps* ($\hat{\cdot}$), i.e. in deviation from their steady-state ($\bar{\cdot}$) levels.

$$\begin{aligned} \hat{\pi}_{t+1} &= \pi_{t+1} - \bar{\pi}_{t+1}, & \hat{y}_{t+1} &= y_{t+1} - \bar{y}_{t+1}, & \hat{\phi}_{t+1} &= \phi_{t+1} - \bar{\phi}_{t+1}, \\ \hat{S}_{t+1} &= \Delta_{8q} S_{t+1} - \Delta \bar{S}_{t+1}, & \hat{r}_{t+1}^f &= r_{t+1}^f - \bar{r}_{t+1}^f, & \hat{\mu}_t &= \mu_t - \bar{\mu}_t. \end{aligned}$$

$\bar{\pi}$ denotes the inflation target, while π_{t+1} refers to the one quarter ahead forecast of four-quarter growth rate in CPI-ATE. See [Hagelund et al. \(2018\)](#) for a detailed description of Norges Bank's approach to estimating the output gap, $\hat{y}$, and potential output, $\bar{y}$. ϕ_t is the four-quarter growth in unit labor cost.⁵ To preserve a stable labor income share, steady-state domestic

³All variables are at a quarterly frequency consistent with the NEMO model, which is a quarterly DSGE model used for monetary policy analysis at Norges Bank.

⁴ λ_Z is estimated to be 0.75.

⁵Unit labor cost is calculated as annual wage growth (from table 3 in MPR 4/25) adjusted for the underlying trend in labor productivity growth (from table 2.1 in MPR 4/25), interpolated by cubic spline to quarterly numbers.

inflation is assumed to coincide with the growth rate of unit labor cost. Under this assumption, $\bar{\phi}$ directly reflects the steady state of domestic inflation⁶. The nominal exchange rate enters the rule as eight-quarter growth, calculated as $(x_t - x_{t-8})/x_{t-8} \times 100$, capturing the medium run movements in the exchange rate. The nominal I-44 index⁷ is treated as a random walk in the forecasting horizon, which implies a steady-state value of $\Delta \bar{S} = 0$. The steady state policy rate $\bar{r}$ for Norway is consistent with Norges Bank’s communicated neutral-rate interval.⁸ The foreign steady state policy rate ($\bar{r}^f$) is estimated to be below $\bar{r}$, consistent with Norges Bank’s assessment of a steady-state risk premium on Norwegian kroner.⁹ The money-market premium μ_t is defined as the spread between three-month NIBOR and the expected average policy rate over the same horizon, and reflects funding and liquidity premiums that vary with international market conditions and FX swap bases. $\bar{\mu}$ is the estimated steady-state money market premium.

The loss function aims to minimize deviations in inflation from target and fluctuations in the output gap, and to smooth policy rate changes over time. The selection of variables in GEORG aims to incorporate macroeconomic indicators of inflation and output in order to implement a forward-looking policy rule. As such, the right-hand side of GEORG should not be interpreted as stabilization targets for the contemporaneous policy rate r_t , but rather as providing a signal of expected inflation and output, which r_t aims to stabilize. Unit labor cost includes both wage and productivity developments. Wage costs represent a large share of marginal costs for domestic firms and can serve as an early indicator of future inflationary pressure. Higher wages, all else equal, will also increase purchasing power and are likely to boost household demand. An increase in productivity decreases the unit labor cost for a given wage level, mitigating inflationary pressure. By explicitly responding to deviations in unit labor cost growth from its steady-state level, monetary policy only responds to wage increases/decreases over and above changes in productivity.

In addition, the krone exchange rate contains useful information about the economic outlook, as it affects inflation and activity through import prices and net exports with a considerable transmission lag. Incorporating foreign policy rates provides information about international economic developments and global monetary and financial developments. In addition, we directly include inflation and the output gap to capture all residual information regarding the economic and inflation outlook not captured by the other indicators. Examples include Norges Bank’s import price indicators (such as IPI and IPK, see for instance [Brubakk et al. \(2024\)](#) and [Norges Bank \(2004\)](#)), price markups, and labor market developments. These variables are not currently included in GEORG as they are not explicitly represented in NEMO. However, efforts are ongoing to integrate them into Norges Bank’s forthcoming model revision.

The set of coefficients in the rule, $\Omega = \{\omega_r, \omega_\pi, \omega_y, \omega_\phi, \omega_S, \omega_{rf}, \omega_\mu\}$, is estimated using IRF matching, see Table 2. See Appendix B for more information about the estimation.

⁶The steady-state of domestic inflation is assumed to be 2.4

⁷The I-44 index ($I44_t$) is a nominal effective exchange-rate index; a higher value indicates a weaker krone.

⁸The latest assessment of the interval was provided in MPR 2/2025, see also [Almlid and Asshoff \(2025\)](#). In NEMO, we need a point estimate of $\bar{r}$, which is within Norges Bank’s published interval.

⁹See Norges Bank (2025) *Monetary Policy Report 2/25*, box “Neutral rate and international interest differentials.”

Table 1: Steady-state values

Variable	Steady-state value
Policy rate ($\bar{r}$)	3.05%
Inflation target ($\bar{\pi}$)	2.0%
Domestic output ($\bar{y}$)	Potential output level
Unit-labor-cost growth ($\bar{\phi}$)	2.4%
Exchange rate change ($\bar{S}$)	0%
Foreign policy rate ($\bar{r}^f$)	2.65%
Money-market premium ($\bar{\mu}$)	0.25 pp

Table 2: Estimated coefficients Ω

Parameter	Estimate
ω_r	0.74
ω_π	1.17
ω_y	1.27
ω_ϕ	1.25
ω_S	0.13
ω_{r^f}	0.25
ω_μ	-1.00

3 Use of GEORG in practice

3.1 The model-based policy rate path

This section compares the policy rate path implied by NEMO with GEORG with the loss function derived policy path. The objective is to assess how closely GEORG replicates the loss function-based policy rate in NEMO. For the *Monetary Policy Report* (MPR) 4/2025, the policy rate paths generated by GEORG and the loss function are nearly identical, and the resulting developments in the output gap, inflation and the exchange rate are similar in the two versions of NEMO.¹⁰ A similar pattern is observed for MPR 3/2025, MPR 2/2025 and MPR 1/2025.

¹⁰Appendix A documents the model-based paths from MPR 3/2025, MPR 2/2025 and MPR 1/2025.

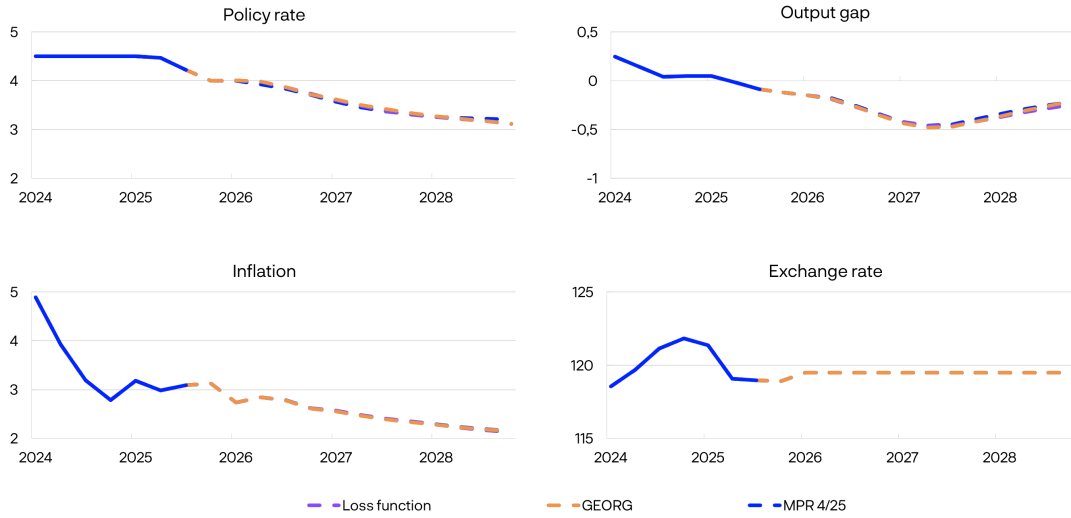


Figure 1: Comparison of NEMO with the loss function and NEMO with GEORG

Another way of assessing the similarities between GEORG and the loss function is to examine the ex-post historical monetary policy shocks. The two series are highly correlated and evolve closely over time, indicating that the rule-based representation captures much of the same systematic behavior embedded in the loss function policy rate.¹¹

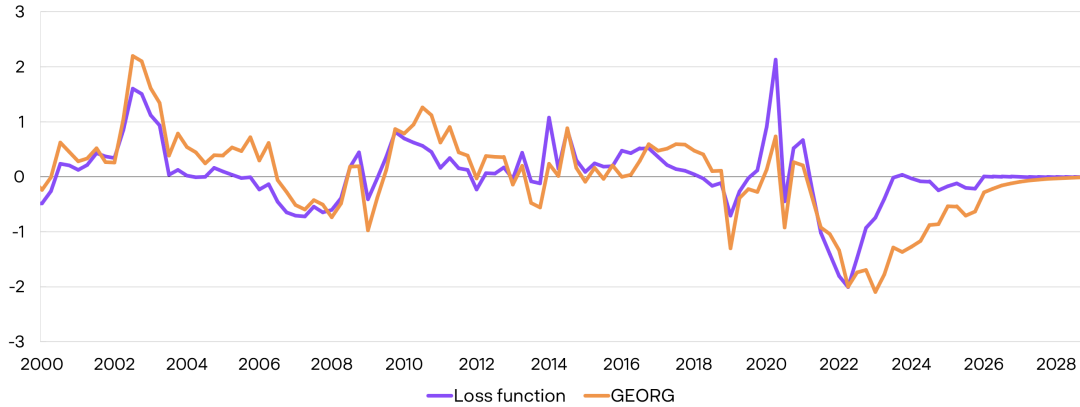


Figure 2: Monetary policy shock

3.2 Decomposition analysis

In the MPR, shock decompositions of the model's policy rate paths are used to identify the factors driving the change in the policy rate forecasts since the previous report. We can also produce a shock decomposition when using NEMO with GEORG. In addition, GEORG allows for a decomposition based on forecast revisions of the observable variables in the rule (or changes in their steady-state values). This endogenous decomposition analysis translates forecast revisions into changes in the policy rate contributions without relying on shock identification. Although this endogenous decomposition is reduced form, it is transparent and can be replicated without detailed knowledge of the model's structure.

¹¹The monetary policy shock in the loss function has no clear structural interpretation and the magnitude of the shock itself is not directly economically meaningful. The relevant comparison lies in the correlation of the two series, which shows that GEORG closely tracks the policy stance implied by the loss function.

In the following, we construct endogenous decomposition analyses using GEORG based on forecast revisions to observable variables and steady-state values for the first three quarters of 2025.

If we denote the GEORG policy rate path at time t as $r_t(\theta_t)$ with $\theta_t = \{\hat{\Omega}, \bar{r}_t\}$ the decomposition from period $t-1$ to t is:

$$\Delta_{t,t-1}r_{t+s} = \mathbb{E}_t[r_{t+s}(\theta_{t+s})] - \mathbb{E}_{t-1}[r_{t+s}(\theta_{t+s})]$$

By defining the change in the forecast of a variable X as:

$$\Delta_{t,t-1}X_{t+s} = \mathbb{E}_t[X_{t+s}] - \mathbb{E}_{t-1}[X_{t+s}]$$

we can decompose the change in the GEORG policy rate path as changes in the forecasts and trend values between the two periods:

$$\begin{aligned} \Delta_{t,t-1}r_{t+s} = & \underbrace{(1 - \omega_r)\omega_\pi \Delta_{t,t-1}\hat{\pi}_{t+1+s}}_{\text{Inflation}} + \underbrace{(1 - \omega_r)\omega_y \Delta_{t,t-1}\hat{y}_{t+1+s}}_{\text{Output gap}} + \underbrace{(1 - \omega_r)\omega_\phi \Delta_{t,t-1}\hat{\phi}_{t+1+s}}_{\text{Unit labor cost}} \\ & + \underbrace{(1 - \omega_r)\omega_s \Delta_{t,t-1}[\Delta_{8q}\hat{s}_{t+1+s}]}_{\text{Exchange rate}} + \underbrace{(1 - \omega_r)\omega_{rf} \Delta_{t,t-1}\hat{r}_{t+1+s}^f}_{\text{Foreign policy rate}} \\ & + \underbrace{(1 - \omega_r)\omega_\mu \Delta_{t,t-1}\hat{\mu}_{t+s}}_{\text{Money market premium}} + \underbrace{\Delta_{t,t-1}\bar{r}_{t+s}}_{\bar{r}} + \underbrace{\omega_r \Delta_{t,t-1}(r_{t-1+s} - \bar{r}_t)}_{\text{Smoothing term}}. \end{aligned} \quad (4)$$

Even though the contributions are calculated mechanically using the rule's coefficients, there are some elements that could benefit from further explanation.

The first is the absence of change in the residual (monetary policy shock), Z_t . Z_t appears in equation (1) but does not appear in equation (4) above. The decomposition analysis aims to illustrate how the model-based path has changed based on changes in the main drivers in the model, and we therefore ignore the residual term. This is also the case for the decompositions published in the Monetary Policy Report.

It can also be useful to go into more detail on how a shift in $\bar{r}$ would enter the decomposition. An upward revision to the point estimate of $\bar{r}$ in NEMO, all else equal, affects the policy path in GEORG through two channels. First, it raises the level term as seen in equation (1) and (4), which shifts the whole model-based path one-to-one. Second, it also enters the policy rate smoothing term $\omega_r(r_{t-1} - \bar{r}_t)$, which partly offsets the level effect. As a result, since $\omega_r > 0$, the policy rate from GEORG does not adjust one-to-one with the revision to $\bar{r}$ in the short run. Figure 3 illustrates these two effects and the resulting net contribution in the decomposition due to a 25 basis points upward revision to $\bar{r}$, all else equal.¹²

¹²Changes in the steady-state money market premium will enter GEORG in the same way as changes in $\bar{r}$.

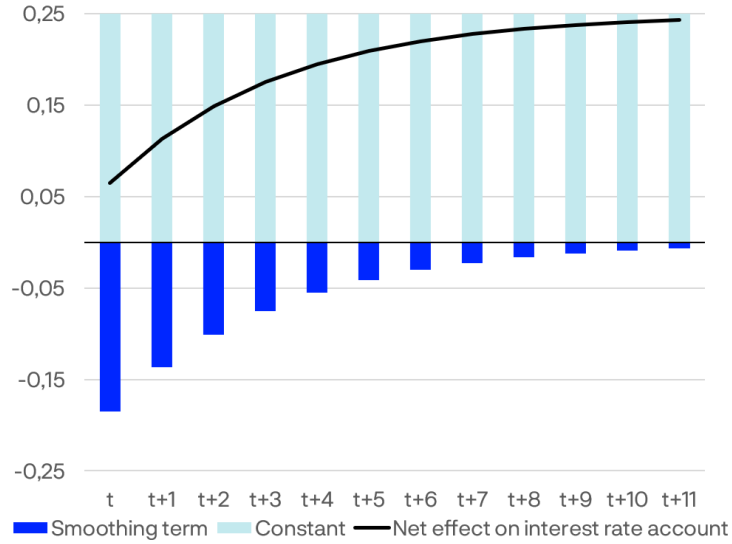


Figure 3: Effect of a change in $\bar{r}$

The policy rate smoothing term will also affect how changes in forecasts over time enter the decomposition. If the inflation forecast is lifted by one percentage point over the entire forecast horizon, the size of the bars in the decomposition will gradually converge to the long-term coefficient of the variable. This is illustrated in figure 4.

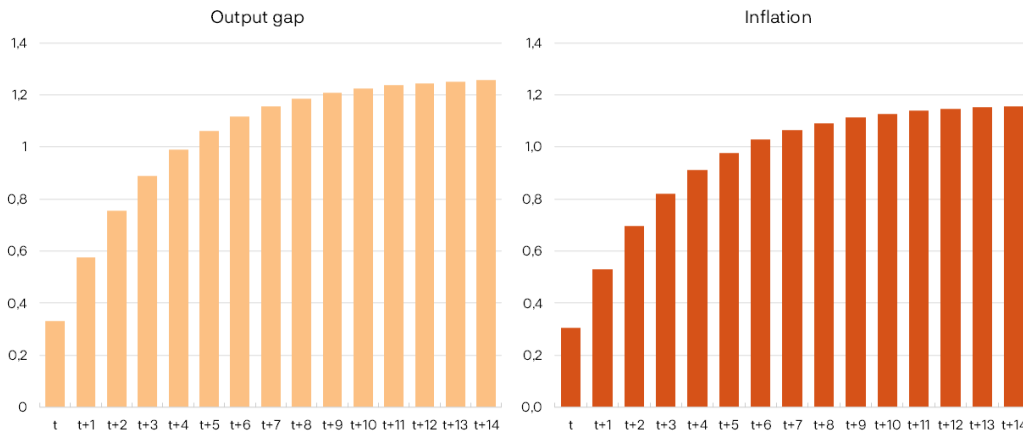


Figure 4: Model-based path response to a 1pp increase in the output gap and inflation forecasts

Figure 5 show the decompositions based on NEMO with GEORG for MPR 4/25. In the forecast based decomposition, revisions to $\bar{r}$ are typically grouped together with either the money market premium or the foreign interest rates, depending on the source of the revision. Revisions to $\bar{r}$ attributed to international factors are assigned to the foreign interest rate bars, while revisions reflecting changes in the long run money market premium assumption are assigned to the money market premium bars. As already mentioned, the GEORG based decompositions use changes in forecasts, while the decomposition traditionally published in MPR is constructed

from changes in shock contributions. Overall, the aggregate change in the model-based policy rate path from NEMO with GEORG is broadly similar to the loss function approach used in the MPR. This is because the coefficients in GEORG are estimated to match the model dynamics of the loss function version of NEMO. Nonetheless, the shock decomposition and the endogenous decomposition based on NEMO with GEORG differ by definition as they are different objects. In the shock decomposition, exchange rate bars typically represent risk premium shocks and their full general equilibrium effects on the policy rate, including the delayed response of inflation. By contrast, in the GEORG (reduced form) decomposition, exchange rate bars show only the role of revisions to the exchange rate forecasts and the delayed inflation responses are reported separately under inflation. The same distinction applies to prices and wages: the shock decomposition bundles price and wage markup shocks, while the GEORG decomposition reports the isolated contribution of changes in price and wage forecasts.

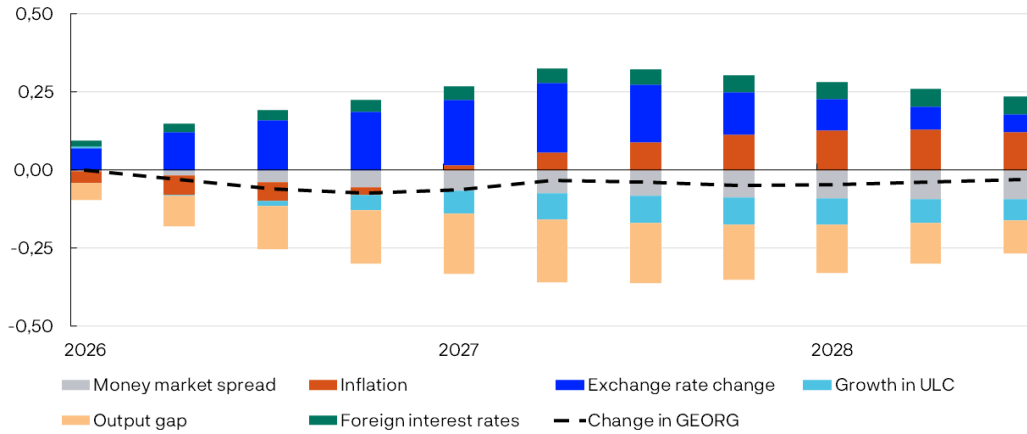


Figure 5: NEMO with GEORG. Decomposition based on changes in projections

4 Conclusion

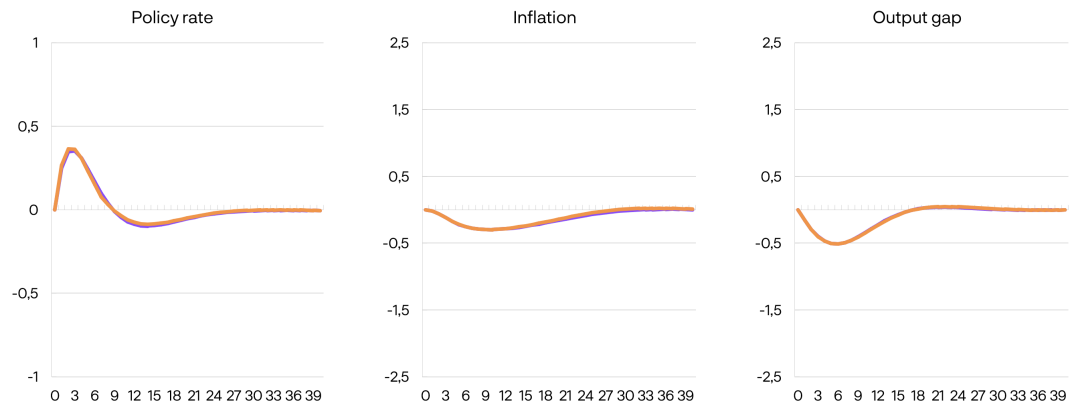
This memo has documented GEORG, a fairly simple, transparent policy rule designed to approximate the loss function policy rate path from Norges Bank’s core model, NEMO. GEORG provides a reduced form mapping of macroeconomic forecasts into an policy rate path. We show that GEORG closely reproduces the dynamics of the loss function approach and can serve as a practical and intuitive tool both for policy analysis and communication. The rule’s primary strength lies in its transparency: it allows both internal and external users to trace how changes in key forecasts affect the implied policy rate path in the model, providing a new endogenous decomposition.

Going forward, we expect GEORG to play a valuable role in Norges Bank’s analytical toolkit. We will continue to develop GEORG going forward, particularly in combination with the development of a new core model in Norges Bank. Improving the treatment of international impulses is a key priority. Further efforts will also explore complementary rules as cross-checks under alternative objectives. These may include rules that are robust across models or that implicitly vary the weight on inflation stabilization relative to employment. Finally, we plan to examine asymmetric reaction functions consistent with Norges Bank’s monetary policy strategy.

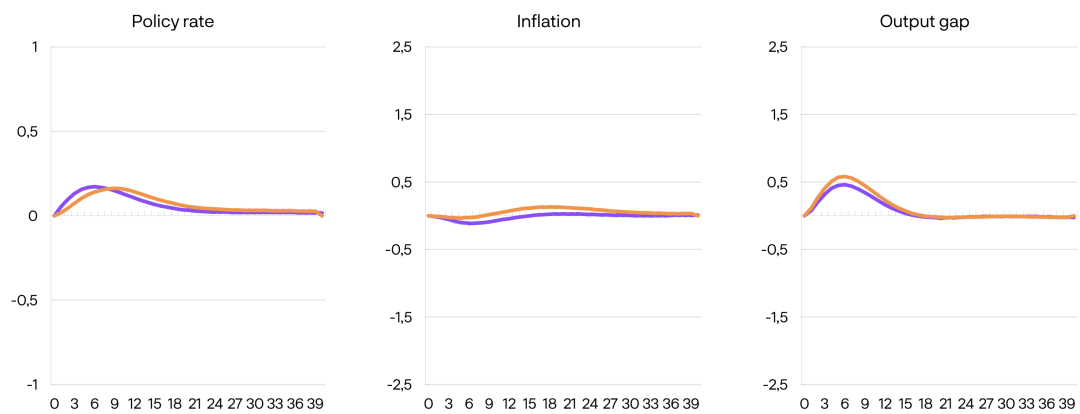
A Appendix: Additional IRFs and account charts

— Loss function — GEORG

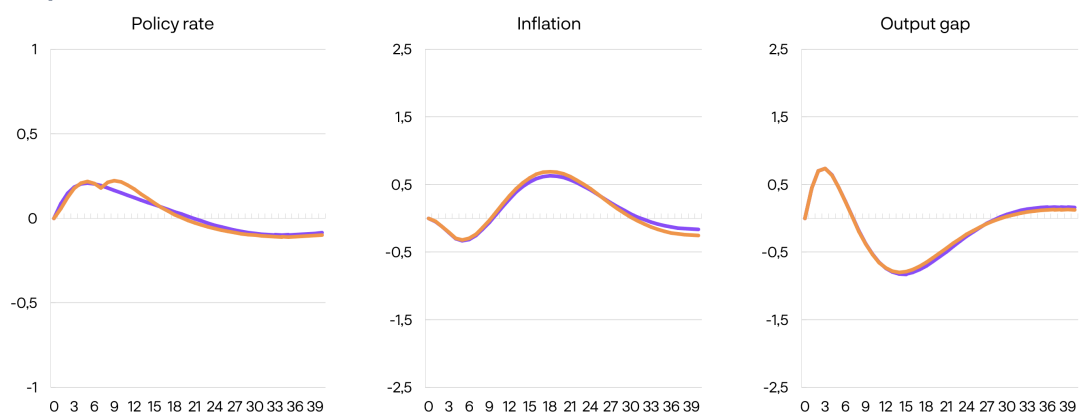
Monetary policy shock



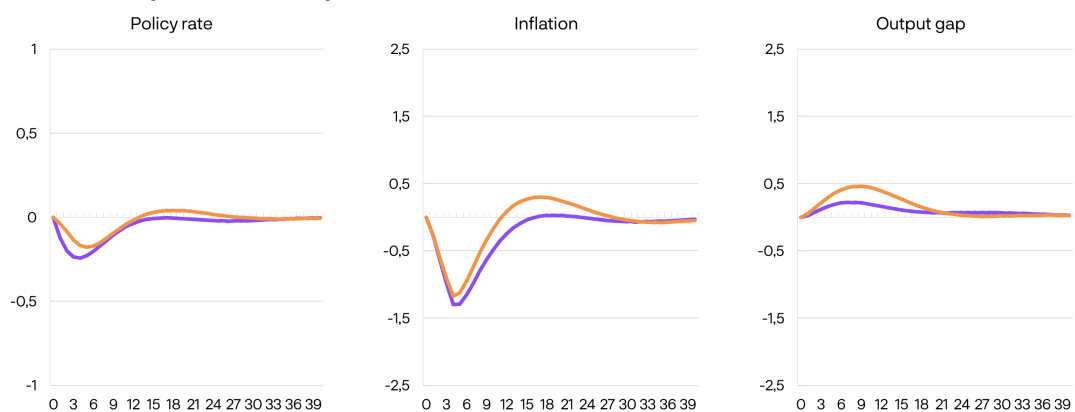
Housing price shock



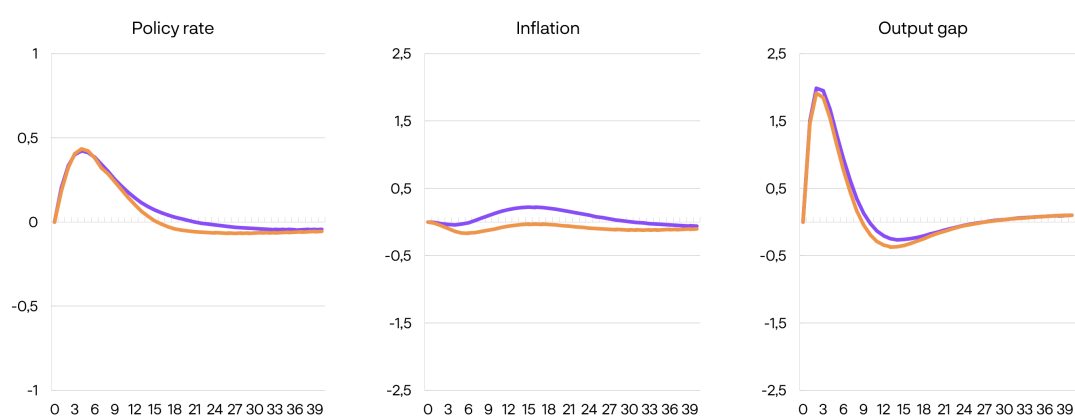
Oil price shock



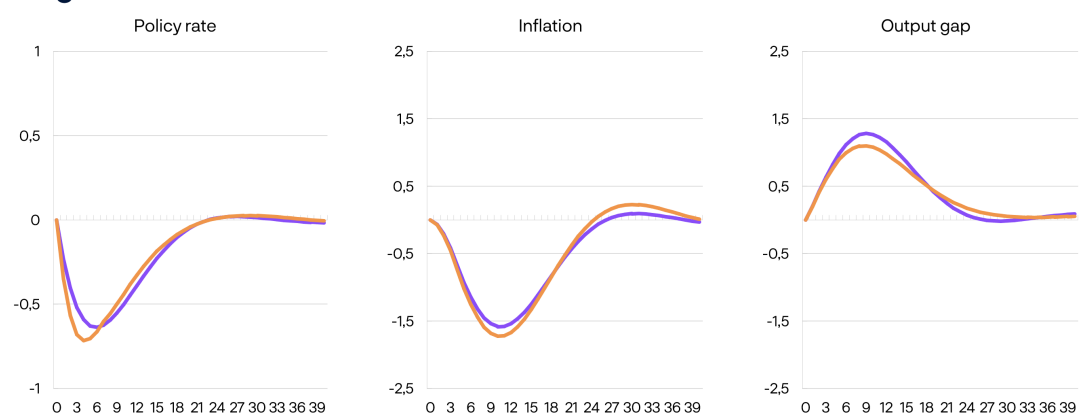
Domestic price markup shock



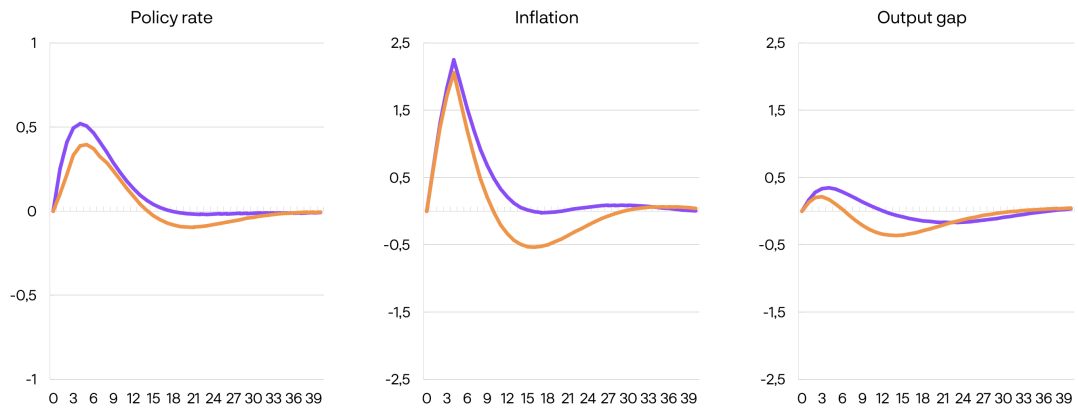
Production shock abroad



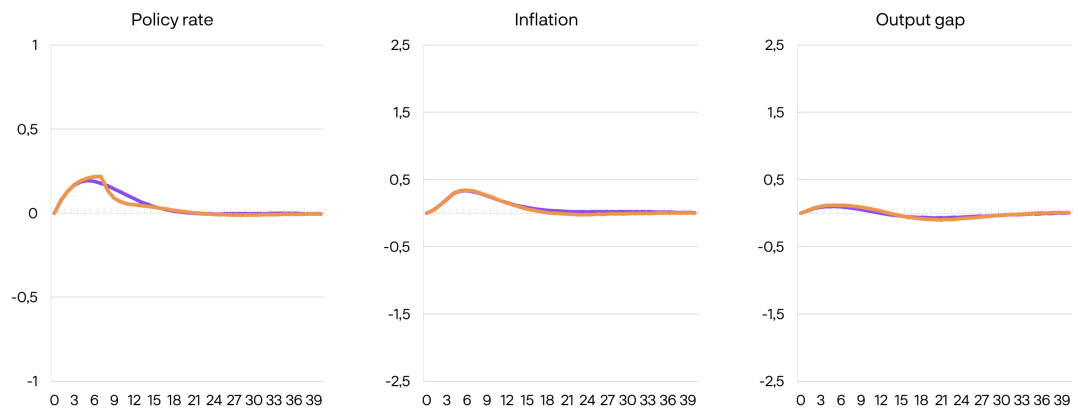
Wage shock



Imported inflation shock



Risk premia shock



Saving shock

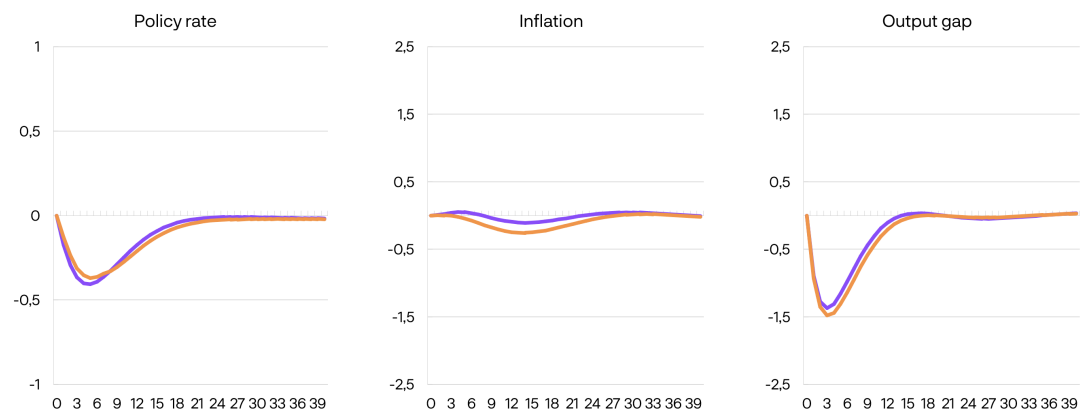


Figure 6: IRF matching

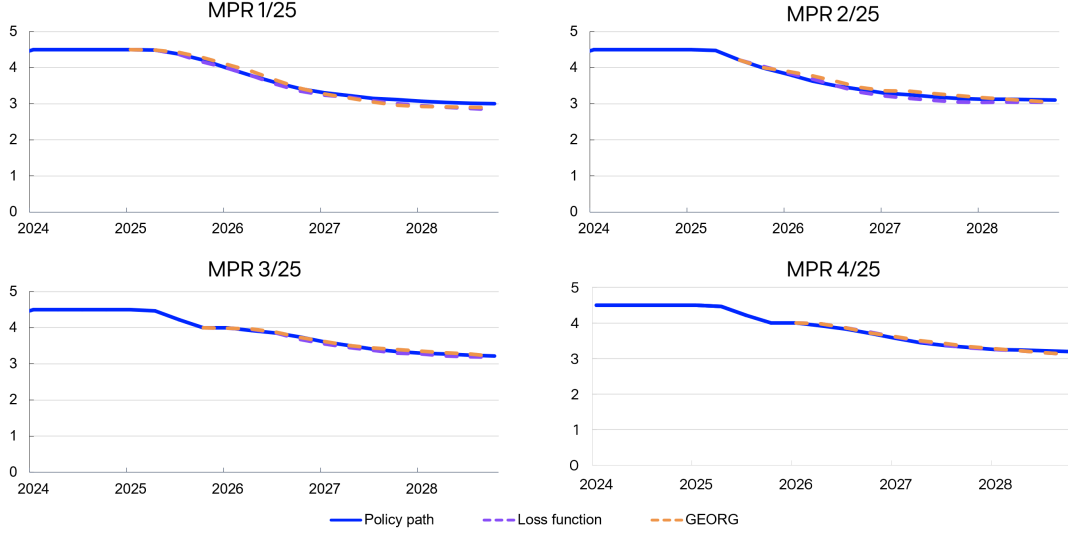


Figure 7: Model-based policy rate paths from the monetary policy reports 2025

B Estimation of coefficients

We estimate $\Omega = \{\omega_r, \omega_\pi, \omega_y, \omega_\phi, \omega_s, \omega_{r,for}, \omega_\mu\}$ such that the dynamic responses for GEORG match those for the benchmark optimal policy (loss function) in NEMO (Kravik and Mimir, 2019). The impulse responses change as we vary the coefficients because economic agents are forward-looking and internalize the central bank’s reaction function. When the parameters in the rule change, the expected policy response changes as well, which in turn alters expectations and the propagation of shocks. By searching over alternative parameter values, we identify the set of coefficients that generates model dynamics most closely aligned with the optimal policy in NEMO.

Consider a set of shocks $\mathcal{S}$ and a vector of model variables $\mathcal{V}$.¹³ Let $\text{IRF}_{v,s}^{(1)}(h)$ denote the h -step impulse response in model 1 (loss-function policy) and $\text{IRF}_{v,s}^{(2)}(h; \Omega)$ the response in model 2 (GEORG). We minimize the weighted quadratic distance:

$$\min_{\Omega \in \mathcal{P}} J(\Omega) = \sum_{s \in \mathcal{S}} \sum_{v \in \mathcal{V}} \sum_{h=0}^H w_{v,s,h} [\text{IRF}_{v,s}^{(2)}(h; \Omega) - \text{IRF}_{v,s}^{(1)}(h)]^2, \quad (5)$$

where $w_{v,s,h} \geq 0$ are optional weights¹⁴, h is the IRF horizon, and $\mathcal{P}$ is the parameter search region implied by our priors.

We use uniform priors for each parameter, which in practice define a search area for the optimizer. The starting value, lower bound and upper bound are given in Table 3.

¹⁵

¹³We match the entire dynamic response of a rich set of variables, not only the policy rate. Both domestic and international shocks are used.

¹⁴We weight the policy rate response relatively more.

¹⁵Uniform priors are natural for IRF matching where identification primarily comes from dynamic fit rather than likelihood curvature. They also avoid unnecessarily weighting at one region and map directly to a search box in numerical optimization.

Table 3: Uniform priors

Parameter	Start	Lower	Upper
ω_r	0.70	0.00	0.99
ω_π	0.32	0.00	3.00
ω_y	0.50	0.00	3.00
ω_ϕ	0.80	0.00	3.00
ω_S	0.10	0.00	3.00
ω_{r^f}	0.26	0.25	3.00
ω_μ	-1.00	-1.01	-0.99

We solve (5) using a population-based metaheuristic from the *artificial bee colony* (ABC) family.¹⁶ The algorithm iteratively evaluates a large number of candidate parameter sets Ω within the admissible search space $\mathcal{P}$. Each candidate is assessed by its objective value $J(\Omega)$, and the population gradually converges toward regions with lower loss. The algorithm returns the parameter vector corresponding to the estimated mode of the objective function.

Table 4 records the estimated annualized coefficients Ω selected by the IRF-matching criterion.

Table 4: Estimated coefficients Ω

Parameter	Estimate
ω_r	0.74
ω_π	1.17
ω_y	1.27
ω_ϕ	1.25
ω_S	0.13
ω_{r^f}	0.25
ω_μ	-1.00

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¹⁶Bee search algorithms use employed, onlooker and scout phases to balance exploration and exploitation, see (Karaboga, 2005; Nozohour-leilabady and Sadeghi, 2016).

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